



UNIVERSITÀ DEGLI STUDI DI MILANO
FACOLTÀ DI SCIENZE E TECNOLOGIE

Categorical Galois Theorem and applications to monoid extensions

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Pullback functor

Let $\sigma : S \rightarrow R$ be a morphism in a category $\mathcal{A}$ with pullbacks.
The functor

$$\begin{aligned}\sigma^* : \mathcal{A}/R &\longrightarrow \mathcal{A}/S \\ a : A \rightarrow R &\longrightarrow p_S : S \times_R A \rightarrow S\end{aligned}$$

has a left adjoint functor

$$\begin{aligned}\Sigma_\sigma : \mathcal{A}/S &\longrightarrow \mathcal{A}/R \\ a : A \rightarrow S &\longrightarrow \sigma \circ a : A \rightarrow R.\end{aligned}$$

Induced adjunction

Let $\mathcal{A} \begin{matrix} \xrightarrow{\mathcal{S}} \\ \perp \\ \xleftarrow{\mathcal{C}} \end{matrix} \mathcal{P}$ be an adjunction where $\mathcal{A}$ has pullbacks.

For every object $S \in \mathcal{A}$, the functor

$$\begin{aligned} \mathcal{S}_S : \mathcal{A}/S &\longrightarrow \mathcal{P}/\mathcal{S}(S) \\ a : A \rightarrow S &\longrightarrow \mathcal{S}(a) : \mathcal{S}(A) \rightarrow \mathcal{S}(S) \end{aligned}$$

has a right adjoint functor

$$\begin{aligned} \mathcal{C}_S : \mathcal{P}/\mathcal{S}(S) &\longrightarrow \mathcal{A}/S \\ f : X \rightarrow \mathcal{S}(S) &\longrightarrow \eta_S^*(\mathcal{C}(f)) : S \times_{\mathcal{C}\mathcal{S}(S)} \mathcal{C}(X) \rightarrow S \end{aligned}$$

where η_S is the unit of $\mathcal{S} \dashv \mathcal{C}$.

Definition

A relatively admissible adjunction $(\mathcal{A}, \overline{\mathcal{A}}) \begin{array}{c} \xrightarrow{\mathcal{S}} \\ \perp \\ \xleftarrow{\mathcal{C}} \end{array} (\mathcal{P}, \overline{\mathcal{P}})$ consists in an adjunction $\mathcal{S} \dashv \mathcal{C}$ and two classes of arrows $\overline{\mathcal{A}}, \overline{\mathcal{P}}$ such that:

- 1 $\overline{\mathcal{A}}$ and $\overline{\mathcal{P}}$ contain all isomorphisms;
- 2 $\overline{\mathcal{A}}$ and $\overline{\mathcal{P}}$ are closed under composition;
- 3 $\overline{\mathcal{A}}$ and $\overline{\mathcal{P}}$ are pullback-stable;
- 4 $\mathcal{S}(\overline{\mathcal{A}}) \subset \overline{\mathcal{P}}$;
- 5 $\mathcal{C}(\overline{\mathcal{P}}) \subset \overline{\mathcal{A}}$;
- 6 $\eta_A \in \overline{\mathcal{A}}$, for every A in $\mathcal{A}$;
- 7 $\varepsilon_X \in \overline{\mathcal{P}}$, for every X in $\mathcal{P}$.

Definition

$a : A \rightarrow R$ is split by $\sigma : S \rightarrow R$ in $\overline{\mathcal{A}}$ when

$$\eta_{\sigma^*(a)}^S : \sigma^*(a) \longrightarrow \mathcal{C}_S \mathcal{S}_S(\sigma^*(a))$$

is an isomorphism.

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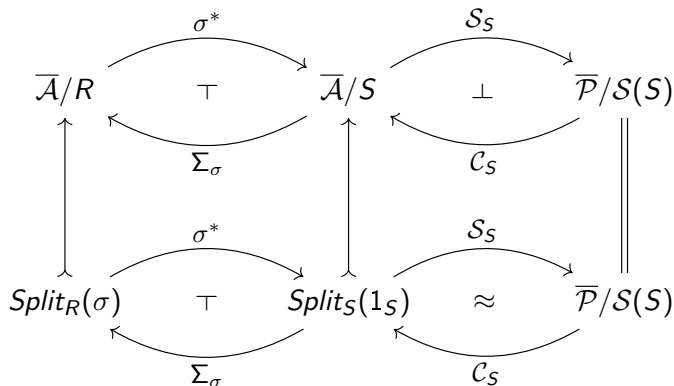
is an isomorphism.

Definition

A morphism $\sigma : S \rightarrow R$ in $\overline{\mathcal{A}}$ is of relative Galois descent with respect to a relatively admissible adjunction when

- ① $\sigma^* : \overline{\mathcal{A}}/R \rightarrow \overline{\mathcal{A}}/S$ is monadic;
- ② $\varepsilon_f^S : f \rightarrow \mathcal{S}_S \mathcal{C}_S(f)$ is an isomorphism, $\forall f : X \rightarrow \mathcal{S}(S)$;
- ③ $\sigma \circ \mathcal{C}_S(f) \in \text{Split}_R(\sigma)$, $\forall f : X \rightarrow \mathcal{S}(S)$.

General setting



$S_S \circ \sigma^* : Split_R(\sigma) \longrightarrow \overline{\mathcal{P}}/S(S)$ is monadic.

The functorial part of the monad is the composite:

$$T^\sigma = S_S \circ \sigma^* \circ \Sigma_\sigma \circ \mathcal{C}_S.$$

Kernel pair groupoid

Consider the kernel pair $S \times_R S \rightrightarrows S \xrightarrow{\sigma} R$ of σ .

It induces an internal groupoid $K(\sigma)$ in $\overline{\mathcal{A}}$:

$$(S \times_R S) \times_S (S \times_R S) \xrightarrow{m} S \times_R S \begin{array}{c} \xleftarrow{\Delta} S \\ \xleftarrow{p_2} S \\ \xleftarrow{p_1} S \end{array}$$

τ (curved arrow from $S \times_R S$ to itself)

fixing:

- ① $\Delta := \langle 1_S, 1_S \rangle$;
- ② $\tau := \langle p_2, p_1 \rangle$;
- ③ $m := \langle \pi_1, \pi_4 \rangle$.

The functor $\mathcal{S}$ transforms $K(\sigma)$ into an internal groupoid in $\overline{\mathcal{P}}$:

$$\begin{array}{ccccc}
 & & & \mathcal{S}(p_1) & \\
 & & & \curvearrowright & \\
 \mathcal{S}(S \times_R S) \times_{\mathcal{S}(S)} \mathcal{S}(S \times_R S) & \xrightarrow{\mathcal{S}(m)} & \mathcal{S}(S \times_R S) & \xleftarrow{\mathcal{S}(\Delta)} & \mathcal{S}(S) \\
 & & \mathcal{S}(\tau) \curvearrowleft & \curvearrowright \mathcal{S}(p_2) & \\
 & & & \mathcal{S}(\tau) &
 \end{array}$$

Definition

The image $\text{Gal}[\sigma] := \mathcal{S}(K(\sigma))$ of $K(\sigma)$ through $\mathcal{S}$ is called the Galois groupoid of σ .

Internal covariant presheaves

An internal covariant presheaf on $Gal[\sigma]$ in $\overline{\mathcal{P}}$ is a triple (X, f, π) where

① $f : X \longrightarrow \mathcal{S}(S)$

② $\pi : \mathcal{S}(S \times_R S) \times_{\mathcal{S}(S)} X \longrightarrow X$

with compatibility with the structure of $Gal[\sigma]$.

The forgetful functor

$$U : \overline{\mathcal{P}}^{Gal[\sigma]} \longrightarrow \overline{\mathcal{P}}/\mathcal{S}(S)$$

$$(X, f, \pi) \longrightarrow f : X \rightarrow \mathcal{S}(S)$$

is monadic.

The functorial part of the monad is the composite:

$$T = \Sigma_{\mathcal{S}(p_2)} \circ \mathcal{S}(p_1)^*.$$

Theorem

Let $(\mathcal{A}, \overline{\mathcal{A}}) \begin{array}{c} \xrightarrow{S} \\ \perp \\ \xleftarrow{C} \end{array} (\mathcal{P}, \overline{\mathcal{P}})$ be a relatively admissible adjunction

where $\mathcal{A}$ and $\mathcal{P}$ are categories with pullbacks. Let $\sigma : S \rightarrow R$ be of relative Galois descent with respect to these data. There exists an equivalence of categories

$$Split_R(\sigma) \approx \overline{\mathcal{P}}^{Gal[\sigma]}$$

between the category of those objects in $\overline{\mathcal{A}}/R$ that are split by σ and the category of internal covariant presheaves on the internal groupoid $Gal[\sigma]$ in $\overline{\mathcal{P}}$.

Classical Galois correspondence

Fix a finite Galois extension $\sigma : K \hookrightarrow L$ of fields.

- ① $\text{Split}_K(\sigma) \cong \text{Split}_K(L)$;
- ② $\text{Gal}[\sigma] \cong \text{Aut}_K(L)$;
- ③ Presheaves on $\text{Gal}[\sigma]$ correspond to group actions of $\text{Aut}_K(L)$ on finite sets.

Contravariant isomorphism between the subfields M

$$K \hookrightarrow M \hookrightarrow L$$

and the subgroups

$$\text{Aut}_L(L) \leq \text{Aut}_M(L) \leq \text{Aut}_K(L).$$

Grothendieck group of monoids

Definition

Let $(M, \cdot, 1)$ be a monoid. The Grothendieck group of M is given by

$$Gp(M) = \frac{GpF(M)}{N(M)},$$

where $GpF(M)$ denotes the free group on M and $N(M)$ the normal subgroup generated by elements of the form $[m_1][m_2][m_1 \cdot m_2]^{-1}$.

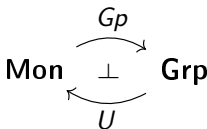
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$$\Gamma = (\mathbf{Mon}, \mathbf{Grp}, Gp, U, \eta, 1_{\mathbf{Grp}}, \mathcal{E}, \mathcal{F})$$

with $\mathcal{E}$ and $\mathcal{F}$ the classes of surjections in $\mathbf{Mon}$ and $\mathbf{Grp}$.

With this choice:

- ① $\mathcal{E}/M \begin{array}{c} \xrightarrow{Gp_M} \\ \perp \\ \xleftarrow{U_M} \end{array} \mathcal{F}/Gp(M)$ is such that U_M is full and faithful;
- ② Every $f \in \mathcal{E}$ is a morphism of descent.

Trivial, central and normal extensions

$a : A \twoheadrightarrow M$ trivial extension when $a \in \text{Split}_M(1_M)$.

$a : A \twoheadrightarrow N$ central extension when $a \in \text{Split}_N(\sigma)$ for some $\sigma \in \mathcal{E}$.

$\sigma : M \twoheadrightarrow N$ normal extension when $\sigma \in \text{Split}_N(\sigma)$.

In the Galois structure $\Gamma = (\mathbf{Mon}, \mathbf{Grp}, Gp, U, \eta, 1_{\mathbf{Grp}}, \mathcal{E}, \mathcal{F})$.

If $\sigma \in \text{Norm}(N)$, there exists an equivalence of categories

$$\text{Centr}(N, \sigma) \approx \mathcal{F}^{\text{Gal}[\sigma]}.$$

Schreier and homogeneous split epimorphisms

Consider a split epimorphism (f, s) of monoid with its kernel:

$$K \hookrightarrow A \begin{array}{c} \xrightarrow{f} \\ \xleftarrow{s} \end{array} M.$$

Definition

(f, s) is a *Schreier split epimorphism* when, for any $a \in A$, there exists a unique $k \in K$ such that $a = k \cdot sf(a)$.

Definition

(f, s) is a *homogeneous split epimorphism* when all the maps

$$\mu_m^r : K \longrightarrow f^{-1}(m), k \longmapsto k \cdot s(m)$$

$$\mu_m^l : K \longrightarrow f^{-1}(m), k \longmapsto s(m) \cdot k$$

are bijective.

Special Schreier and special homogeneous surjections

Consider a surjection $\sigma : M \twoheadrightarrow N$ of monoid with its kernel pair:

$$M \times_N M \begin{array}{c} \xrightarrow{p_1} \\ \xleftarrow{\Delta} \\ \xrightarrow{p_2} \end{array} \twoheadrightarrow M \xrightarrow{\sigma} \twoheadrightarrow N$$

Definition

$\sigma : M \twoheadrightarrow N$ is a *special Schreier surjection* when (p_1, Δ) is a Schreier split epimorphism.

Definition

$\sigma : M \twoheadrightarrow N$ is a *special homogeneous surjection* when (p_1, Δ) is a homogeneous split epimorphism.

Semidirect form of Schreier split epimorphisms

Proposition

There is an isomorphism of Schreier split epimorphisms

$$\begin{array}{ccccccc}
 K & \xrightarrow{\quad} & M \times_N M & \begin{array}{c} \xrightarrow{p_1} \\ \xleftarrow{\Delta} \\ \xrightarrow{p_2} \end{array} & M & \xrightarrow{\sigma} & N \\
 \parallel & & \uparrow \zeta^{-1} \downarrow \zeta & & \parallel & & \parallel \\
 K & \xrightarrow{\quad} & K \rtimes_{\varphi} M & \begin{array}{c} \xrightarrow{\quad} \\ \xleftarrow{\quad} \\ \xrightarrow{\bullet} \end{array} & M & \xrightarrow{\sigma} & N
 \end{array}$$

- ① $\varphi : M \rightarrow \text{End}(K)$ is given by $\varphi(m)(k) \cdot m = m \cdot k$;
- ② $K \rtimes_{\varphi} M$ is the semidirect product with respect to φ ;
- ③ $\zeta(m_1, m_2) = (k, m_2)$, where $m_1 = k \cdot m_2$;
- ④ $\zeta^{-1}(k, m) = (k \cdot m, m)$;
- ⑤ $\bullet(k, m) = k \cdot m$.

Theorem

Consider a split epimorphism (f, s) ; then f is a trivial extension if and only if f is a special homogeneous surjection.

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Theorem

Let $\sigma : M \twoheadrightarrow N$ be a surjective homomorphism of monoids. The following are equivalent:

- ① σ is a central extension;*
- ② σ is a normal extension;*
- ③ σ is a special homogeneous surjection.*

Semidirect form of $K(\sigma)$

Consider $\sigma \in \text{Norm}(N)$ and its kernel pair groupoid $K(\sigma)$

$$(K \rtimes_{\varphi} M) \times_M (K \rtimes_{\varphi} M) \xrightarrow{m} K \rtimes_{\varphi} M \begin{array}{c} \xleftarrow{\quad} M \\ \xrightarrow{\quad} M \end{array}$$

$\begin{array}{c} \text{curved arrow from } K \rtimes_{\varphi} M \text{ to } M \\ \text{curved arrow from } M \text{ to } K \rtimes_{\varphi} M \end{array}$

$\begin{array}{c} \text{curved arrow from } K \rtimes_{\varphi} M \text{ to } K \rtimes_{\varphi} M \text{ labeled } \tau \\ \bullet \end{array}$

where:

- ① $\varphi : M \rightarrow \text{Aut}(K)$ is given by $\varphi(m)(k) \cdot m = m \cdot k$;
- ② $\bullet(k, m) = k \cdot m$;
- ③ $\tau(k, m) = (k^{-1}, k \cdot m)$;
- ④ $m(k_1, m_2, k_2, m_3) = (k_1 \cdot k_2, m_3)$.

Semidirect form of $Gal[\sigma]$

Consider $\sigma \in Norm(N)$ and its Galois groupoid $Gal[\sigma] = Gp(K(\sigma))$

$$\begin{array}{c}
 (K \rtimes_{\overline{\varphi}} Gp(M)) \times_{Gp(M)} (K \rtimes_{\overline{\varphi}} Gp(M)) \xrightarrow{m} K \rtimes_{\overline{\varphi}} Gp(M) \xleftarrow{\quad} Gp(M) \\
 \begin{array}{ccc}
 & \curvearrowright & \\
 \cup \tau & & \bullet \\
 & \curvearrowleft &
 \end{array}
 \end{array}$$

where:

- ① $\overline{\varphi} : Gp(M) \rightarrow Aut(K)$ is given by $\overline{\varphi}(\overline{[m]})(k) \cdot m = m \cdot k$;
- ② $\bullet(k, \overline{[m]}) = \overline{[k]} \cdot \overline{[m]}$;
- ③ $\tau(k, \overline{[m]}) = (k^{-1}, \overline{[k]} \cdot \overline{[m]})$;
- ④ $m(k_1, \overline{[m_2]}, k_2, \overline{[m_3]}) = (k_1 \cdot k_2, \overline{[m_3]})$.

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